

CE 4610 Week 3 - Virginia Statewide Crosswalk Inventory

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13 September 2026

The Virginia Statewide Crosswalk inventory is a collection of all of the crosswalks in Virginia. The crosswalks are identified from Virginia Geographic Information Network (VGIN)'s aerial imagery data. There are 68,859 crosswalks, and the Virginia Department of Transportation (VDOT) states that their count is highly accurate (100% accuracy) for areas of the state with VDOT Linear Referencing System (LRS) available.

Encoding (bytes)

Various text formats for tabular data including markdown languages, comma separated values, and JavaScript Object notation: ArcGIS Hub Dataset (HTML), ArcGIS GeoServices (REST API), CSV, Shapefile, GeoJSON, and KML. Features are saved as strings, integers, and floats as applicable.

Schema (structure)

The dataset is made up of rows (entries) and columns (features). Each row is a crosswalk and each column describes qualities of that crosswalk including Web Mercator coordinates, longitude and latitude, county, city, district, design category, and a unique identification number.

Vocabulary (meaning)

- The primary language of the dataset is geographic location of the crosswalks, but the exact location is expressed using seven of the eleven columns. The crosswalk's position can be found based on the dataset and information about a county/city/district's traffic can also be derived.
- Crosswalks are classified into 6 categories based on the painted pattern: (1) Transverse lines, (2) Double paired, (3) Longitudinal bar, (4) Diagonal, (5) Perpendicular bar, (6) Other.
- Additionally, crosswalks made of brick are indicated with "brick" using a value of "0" (no brick), "1" (brick path in one direction), and "2" (brick path in two directions, like a crosshair intersection).

Transport (delivery)

Hypertext Transfer Protocol Secure (HTTPS): The dataset is available for download on several open-source websites, including a [virginia.gov](https://www.virginia.gov) domain and VDOT website.

<https://www.arcgis.com/home/item.html?id=1a1b51d6c3624a458d6c9d5b40181b5c&sublayer=0#overview>

<https://www.virginiaroads.org/datasets/VDOT::virginia-statewide-crosswalk-inventory/about>

<https://data.virginia.gov/dataset/virginia-statewide-crosswalk-inventory>

Representational State Transfer (REST) API:

https://services.arcgis.com/p5v98VHDX9Atv3I7/ArcGIS/rest/services/Virginia_Statewide_Crosswalk_Inventory/FeatureServer/0

Where do you see metadata for the dataset?

Detailed metadata for the dataset is available at the ArcGIS REST API access point. Minimal information about the dataset owner is available at the HTTPS download locations.